

19. Graphics creation

19.1 Common graphics skills

19.2 Vector graphics

19.3 Bitmap images

19.4 Compression

19.5 Text

Learning Outcomes :

Remember

- Identify common graphics tools, transform tools, layout tools, and colour systems used in digital graphics editing.
- Recall the purpose of layers, alignment tools, crop tools, and colour picker tools.

Understand

- Explain how different graphics tools help create, edit, organize, and improve digital images.
- Describe the difference between bitmap and vector editing operations and the role of colour systems.

Apply

- Use layers, transformations, grouping, alignment, cropping, and colour tools to create and edit digital graphics effectively.
- Apply appropriate layout and colour tools to produce accurate and professional graphic designs.

Analyze

- Compare different graphics editing tools and determine which tool is most suitable for a specific design task.
- Examine how transformations, layer order, and object alignment affect the final appearance of a graphic.

Evaluate

- Assess the effectiveness of different editing techniques, colour systems, and layout tools in achieving design objectives.
- Justify the selection of specific graphics tools for professional digital design work.

Create

- Design and edit digital graphics using a combination of layers, transformations, alignment, colour, and layout tools.
- Produce visually organized and professional-quality digital artwork using appropriate graphics software.

Common Graphics Skills

Digital graphics software uses a variety of tools to create, edit, and organize visual content efficiently. One of the most important concepts in graphics editing is **layers**.

A layer can be understood as a transparent sheet placed above or below other sheets, where each sheet contains a different part of the image. This allows individual elements to be edited independently without affecting the entire design. According to the chapter, layers help move or edit parts of an image separately, making design work more flexible and accurate.

1. Add Layers

Explanation

Adding a layer means creating a new transparent workspace where new objects, text, images, or effects can be placed.

Instead of drawing everything on one surface, designers separate elements into different layers for easier control.

Why it is used

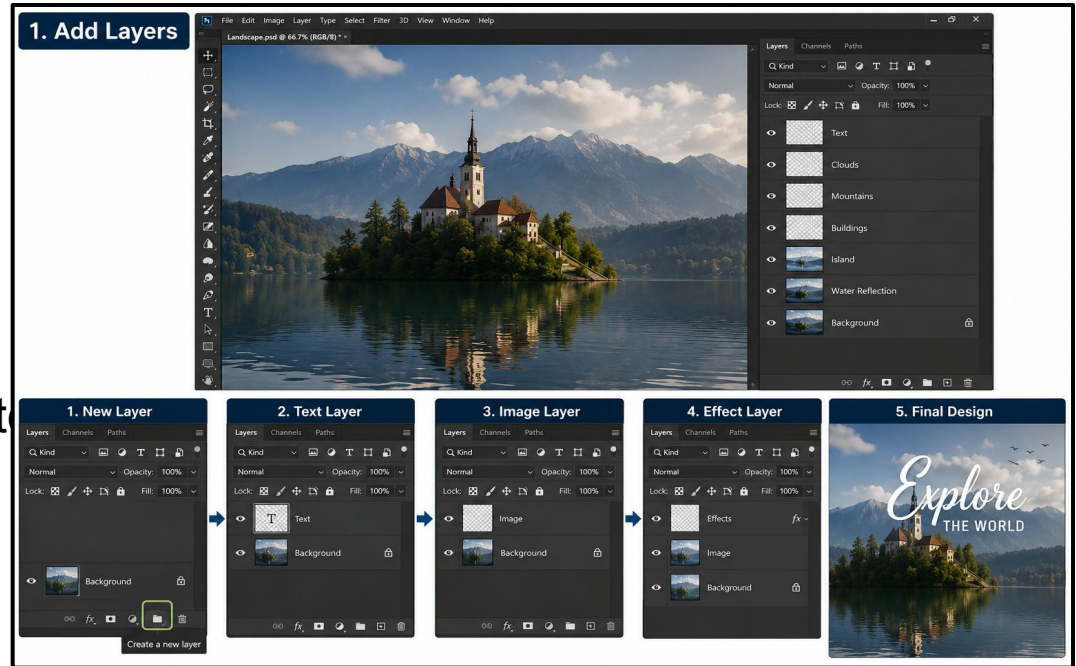
- keeps design organized
- allows independent editing
- prevents accidental changes to other objects
- supports complex multi-element designs

Example

A poster design may contain:

- background image on one layer
- title text on another layer
- logo on another layer
- decorative icons on separate layers

This makes editing easier.



2. Remove Layers

Explanation

Removing a layer deletes unwanted elements from the design.

If an object is no longer needed, its entire layer can be deleted without affecting the remaining design.

Why it is used

- removes mistakes quickly
- cleans unnecessary elements
- simplifies project structure

Example

If the wrong logo was inserted, deleting that specific layer removes only the logo.

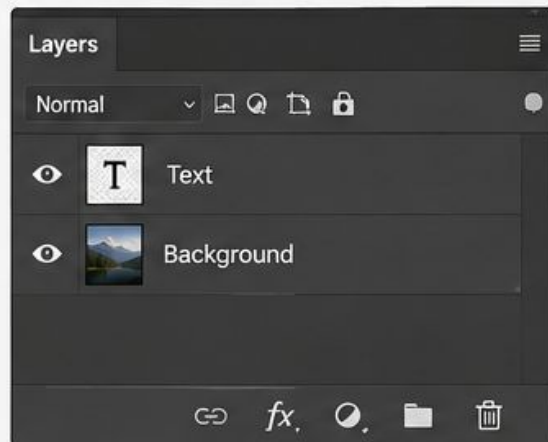
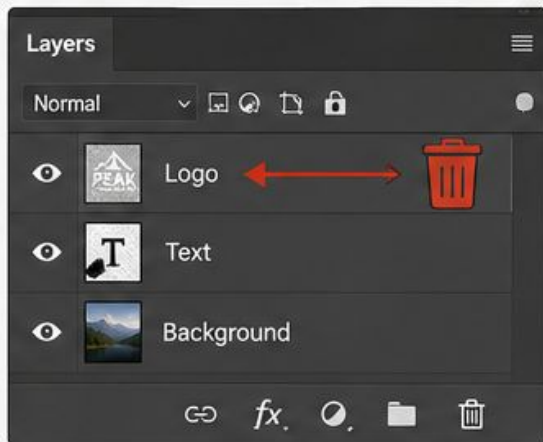
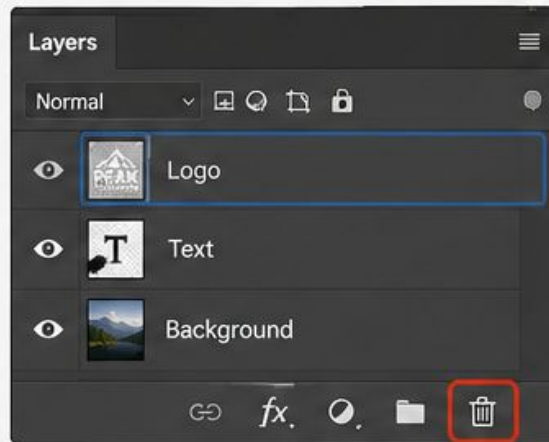
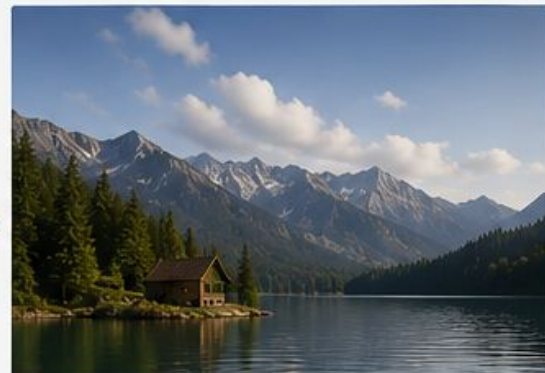
Original Design (With Extra Logo)



Delete Logo Layer



Final Design (Logo Removed)



3. Select Active Layer

The active layer is the currently selected layer where editing actions will occur.

Any drawing, transformation, text insertion, or colour change affects only the active layer.

Why it is important

Without selecting the correct layer:

- edits may affect the wrong object
- unwanted modifications occur
- design workflow becomes confusing

Example

If the text layer is selected:
changing font colour affects text only.

4. Layer Order

Explanation

Layer order determines which object appears in front and which appears behind.

Objects in upper layers cover objects in lower layers.

Example

Imagine:

Layer 1 = background sky

Layer 2 = tree

Layer 3 = bird

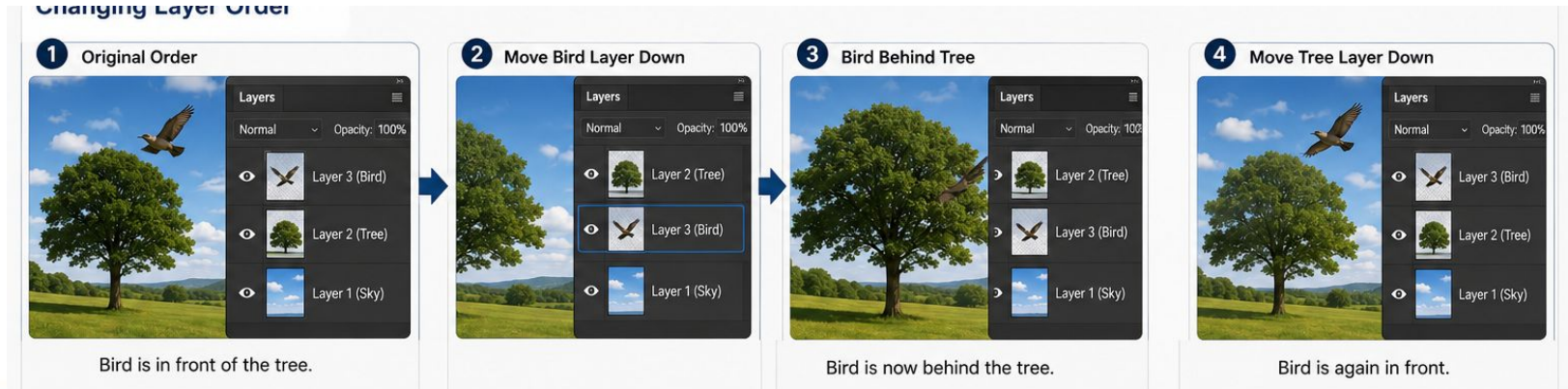
The bird appears in front of the tree because its layer is above.

Why it is used

Controls visual hierarchy and depth.

Useful for:

- poster design
- web banners
- advertisements
- scene compositions



5. Toggle Visibility

Explanation

Toggle visibility allows a layer to be temporarily shown or hidden.

Hidden layers remain in the project but are not visible on screen.

Why it is useful

- simplifies editing
- isolates specific objects
- helps compare versions
- reduces visual clutter

Example

While editing text, background visibility may be turned off for clarity.

6. Lock / Unlock Layers

Locking a layer prevents accidental editing.

When locked:

- objects cannot be moved
- deleted
- painted over
- modified

Unlocking restores editing access.

Why it is useful

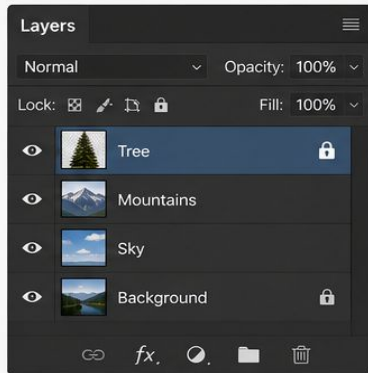
Protects important design elements.

Example

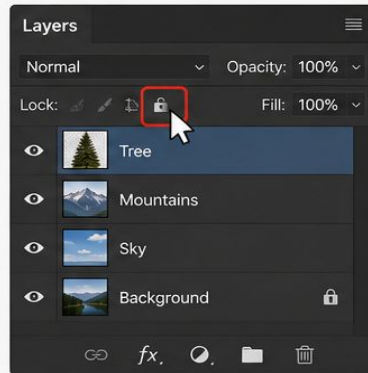
Locking the background image while editing foreground text.



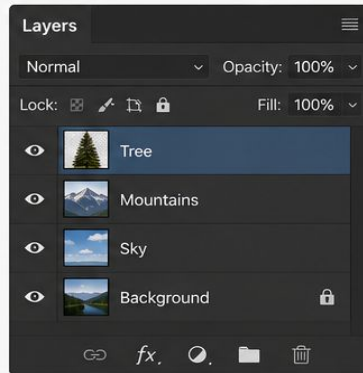
1. Layer Locked



2. Click Lock Icon



3. Layer Unlocked



4. Editing Enabled



7. Change Opacity / Transparency

Explanation

Opacity controls visibility intensity.

100% opacity:
fully visible

50% opacity:
semi-transparent

0% opacity:
invisible

Transparency is the opposite of opacity.

Why it is used

Creates:

- watermark effects
- overlays
- faded backgrounds
- blending effects

Example

A company logo watermark behind text.

8. Blend Layers

Blending mixes colours of one layer with layers beneath it.

Different blend modes create different visual effects.

Examples:

- Multiply
- Overlay
- Screen
- Colour Burn

Examples

- Multiply
- Overlay
- Screen
- Colour Burn

Layers Used

- Texture / Gradient
- Landscape

1. Normal (Default)

Layers Channels Paths

Normal Opacity: 100%

Lock: Fill: 100%

Texture / Gradient

Landscape



Normal (Default)

2. Multiply

Layers Channels Paths

Multiply Opacity: 100%

Lock: Fill: 100%

Texture / Gradient

Landscape



Multiply

3. Overlay

Layers Channels Paths

Overlay Opacity: 100%

Lock: Fill: 100%

Texture / Gradient

Landscape



Overlay

4. Screen

Layers Channels Paths

Screen Opacity: 100%

Lock: Fill: 100%

Texture / Gradient

Landscape



Screen

5. Colour Burn

Layers Channels Paths

Colour Burn Opacity: 100%

Lock: Fill: 100%

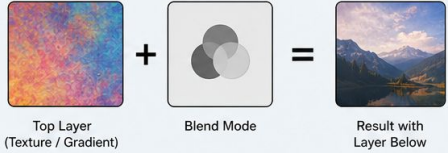
Texture / Gradient

Landscape



Colour Burn

How It Works



Top Layer (Texture / Gradient) + Blend Mode = Result with Layer Below

Why it is used

Creates:

- lighting effects
- shadows
- texture overlays
- artistic designs

Example

Adding paper texture over poster background.

9. Merge Layers

Merge combines selected layers into one single editable layer.

Only chosen layers are merged.

Why it is used

- reduces layer clutter
- combines related objects
- improves organization

Example

Merge logo icon + company name text.

10. Flatten Layers

Explanation

Flatten combines ALL layers into one final image.

After flattening:
individual layers cannot be edited separately.

Why it is used

- final export
- reduce file complexity
- prepare finished image

Example

Flattening completed magazine cover before saving.

Transform Tools in Digital Graphics

1. Resize

Explanation:

Resize changes the width and height of an object to make it larger or smaller. It allows independent adjustment of dimensions.

Why it is used:

- image is too large
- object is too small
- design needs size adjustment

Example:

A logo inserted into a poster appears too large. Resize reduces its dimensions to fit properly.

Real-life use:

Poster design, banner editing, social media graphics.

2. Skew

Explanation:

Skew tilts an object sideways, making it appear slanted.

Why it is used:

- create angled designs
- make text or shapes dynamic
- add creative visual effects

Example:

A rectangle can be skewed to look like a tilted signboard.

Real-life use:

Advertisements, typography design.

3. Reflect

Explanation:

Reflect creates a mirror image of an object either horizontally or vertically.

Why it is used:

- create symmetry
- duplicate mirrored designs
- generate balanced layouts

Example:

Reflecting one butterfly wing to create a full butterfly.

Real-life use:

Logo design, graphic composition.

4. Rotate

Explanation:

Rotate turns an object around a fixed point at a specific angle.

Why it is used:

- correct orientation
- create dynamic layouts
- position icons properly

Example:

Rotating an arrow upward in a presentation.

Real-life use:

Posters, icons, diagrams.

5. Move

Explanation:

Move changes the position of an object without changing its size or shape.

Why it is used:

- reposition elements
- organize layout
- improve alignment

Example:

Moving a company logo to the top-right corner.

Real-life use:

All digital designs.

6. Scale

Explanation:

Scale changes object size proportionally while maintaining original shape.

Why it is used:

- avoid distortion
- resize logos/images accurately

Example:

Scaling a company logo larger while keeping proportions.

Real-life use:

Branding, print design.

7. Shear

Explanation:

Shear distorts an object by shifting parts in different directions.

Why it is used:

- create dramatic shape effects
- stylized transformations

Example:

Converting a square into a parallelogram.

Real-life use:

Creative posters.

8. Apply Envelope

Explanation:

Envelope reshapes objects using adjustable control points or mesh.

Why it is used:

- warp text/images
- fit designs onto curved surfaces

Example:

Wrapping text around a bottle.

Real-life use:

Packaging design.

9. Perspective

Explanation:

Perspective changes viewing angle to create realistic depth.

Why it is used:

- create 3D effects
- realistic mockups

Example:

Placing poster artwork on a billboard image.

Real-life use:

Advertising mockups.

Grouping or Merging Tools

1. Group

Grouping combines multiple objects so they can be selected, moved, resized, or edited together as a single unit without changing their individual design.

Why it is used:

- organize multiple objects
- move several items together
- save editing time
- maintain layout consistency

Example:

Grouping text, icon, and shape in a logo design so they move together.

Real-life use:

Logo design, posters, presentations.

2. Ungroup

Explanation:

Ungroup separates previously grouped objects so each item can be edited individually.

Why it is used:

- modify a specific object
- change colour or size of one element
- edit individual design parts

Example:

Ungrouping a logo to edit only the icon without affecting the text.

Real-life use:

Graphic editing, logo modification.

3. Combine / Join

Combine or Join merges multiple shapes or objects into one single object.

Unlike grouping, the objects become permanently connected as one shape.

Why it is used:

- create custom shapes
- simplify design structure
- make complex objects from simple shapes

Example:

Combining circles and rectangles to create a custom icon.

Real-life use:

Logo creation, icon design.

4. Add / Subtract

These tools modify shapes mathematically.

Add: merges selected shapes into one larger shape.

Subtract: removes one shape from another to create cut-out effects.

Why it is used:

- create unique shapes
- design hollow objects
- produce creative visual effects

Example:

Subtracting a small circle from a rectangle to create a keyhole shape.

Real-life use:

UI icons, branding graphics.

5. Intersect

Intersect keeps only the overlapping portion of two or more shapes and removes the rest.

Why it is used:

- create precise shape combinations
- design custom overlapping graphics
- generate complex visual elements

Example:

Overlapping two circles and keeping only the shared middle area.

Real-life use:

Modern logo design, vector illustrations.

Alignment and Distribution Tools

1. Left Alignment

Left alignment positions multiple objects so that their left edges line up in a straight vertical line.

Why it is used:

- create neat layouts
- maintain consistent positioning
- improve visual organization

Example:

Aligning multiple text boxes along the left side of a poster.

Real-life use:

Posters, presentations, website layouts.

2. Right Alignment

Explanation:

Right alignment arranges objects so that their right edges line up evenly.

Why it is used:

- maintain balanced design
- create organized layouts
- improve readability

Example:

Aligning buttons evenly on the right side of a webpage design.

Real-life use:

UI design, brochures.

3. Top Alignment

Explanation:

Top alignment places objects so that their top edges are positioned at the same level.

Why it is used:

- keep objects visually balanced
- maintain consistent header layouts

Example:

Aligning multiple images in the same row.

Real-life use:

Photo galleries, banners.

4. Bottom Alignment

Explanation:

Bottom alignment arranges objects so their bottom edges line up evenly.

Why it is used:

- maintain symmetry
- improve layout consistency

Example:

Aligning icons at the bottom of an infographic.

Real-life use:

Dashboards, presentations.

5. Centre Alignment

Explanation:

Centre alignment positions objects equally from both sides so they appear perfectly centered.

Why it is used:

- create balanced design
- focus attention on important elements

Example:

Placing a title in the exact center of a poster.

Real-life use:

Certificates, invitations, presentations.

6. Vertical Distribution

Explanation:

Vertical distribution spaces multiple objects evenly from top to bottom.

Why it is used:

- create equal spacing
- improve visual consistency

Example:

Evenly spacing menu buttons vertically.

Real-life use:

Mobile app design, forms.

7. Horizontal Distribution

Explanation:

Horizontal distribution spaces multiple objects evenly from left to right.

Why it is used:

- create clean layouts
- avoid uneven spacing

Example:

Evenly spacing icons across a toolbar.

Real-life use:

Web design, presentations.

8. Order

Order determines the stacking position of objects in a design.

Objects placed above appear in front, while lower objects remain behind.

Why it is used:

- control visibility
- create layered designs
- organize overlapping elements

Example:

Placing text above an image background.

Real-life use:

Posters, advertisements.

9. Raise

Explanation:

Raise moves an object one level upward in the layer order.

Why it is used:

- adjust overlapping objects
- improve object visibility

Example:

Bringing a button slightly above another object.

Real-life use:

UI design.

10. Lower

Explanation:

Lower moves an object one level downward in the layer stack.

Why it is used:

- place objects behind others
- adjust design hierarchy

Example:

Moving decorative shapes behind text.

Real-life use:

Flyers, banners.

11. Bring to Front

Explanation:

Bring to Front moves an object to the topmost layer so it becomes fully visible above all other objects.

Why it is used:

- highlight important content
- solve overlapping visibility issues

Example:

Bringing a company logo above background graphics.

Real-life use:

Advertisements, branding.

12. Send to Back

Explanation:

Send to Back moves an object to the lowest layer behind all other objects.

Why it is used:

- create background elements
- manage design depth

Example:

Sending a background image behind text and icons.

Real-life use:

Poster design, presentations.

Layout Tools

1. Rulers

Rulers are measurement guides displayed along the horizontal and vertical edges of the workspace. They help designers measure object size, spacing, and positioning accurately.

Why it is used:

- measure dimensions precisely
- maintain equal spacing
- position objects accurately
- improve design consistency

Example:

Using rulers to place a logo exactly 2 cm from the top edge of a poster.

Real-life use:

Poster design, brochure layout, web page design.

2. Grids

A grid is a pattern of horizontal and vertical lines displayed in the background of the workspace. It acts as a visual guide for placing and aligning objects.

Grids do not appear in the final printed or exported design.

Why it is used:

- align objects accurately
- maintain equal spacing
- improve design structure
- create balanced layouts

Example:

Using a grid to arrange multiple icons evenly in an infographic.

Real-life use:

UI design, posters, presentations.

3. Guidelines

Guidelines are movable horizontal or vertical reference lines placed at specific positions in the workspace. Unlike grids, guidelines can be positioned manually wherever precise alignment is needed.

Why it is used:

- align objects precisely
- maintain consistency
- control layout structure
- improve placement accuracy

Example:

Placing a vertical guideline at the center of a certificate to align the title perfectly.

Real-life use:

Certificates, magazine layouts, professional graphics.

4. Snapping

Snapping automatically aligns objects to grids, guidelines, or nearby objects when they are moved close to them. This helps objects “snap” into precise positions.

Why it is used:

- improve alignment accuracy
- speed up design work
- avoid uneven placement
- create professional layouts

Example:

When moving a text box, it automatically aligns to the center guideline.

Real-life use:

Graphic design, interface design, document layout.

Colour Picker Tools

1. Select a Colour

Colour picker tools allow designers to select a specific colour for use in a design. Colours can be chosen from the software's colour palette, sliders, or colour wheel.

This helps maintain accurate and consistent colour usage throughout the design.

Why it is used:

- choose exact colours
- maintain design consistency
- match brand colours
- improve visual appeal

Example:

Selecting blue from the colour palette for a company logo.

Real-life use:

Logo design, posters, branding.

2. Select a Colour from an Existing Image

The Eyedropper (Colour Picker) tool allows users to pick a colour directly from an existing image or object. This ensures the exact same shade is reused elsewhere.

Why it is used:

- match existing colours accurately
- maintain brand consistency
- save colour selection time

Example:

Picking the exact red colour from a company logo to use in a brochure design.

Real-life use:

Brand identity design, photo editing, advertisements.

3. Select a Colour Range

Colour range selection allows users to select multiple pixels or areas that contain similar colours within an image.

Instead of selecting one exact colour, a group of related shades can be selected.

Why it is used:

- edit similar colours together
- replace backgrounds
- adjust specific colour regions
- speed up editing

Example:

Selecting all shades of blue in the sky to change them to sunset orange.

Real-life use:

Photo editing, image correction, background replacement.

Crop Tools

1. Crop

Crop removes unwanted outer portions of an image while keeping only the required area. It improves composition and focuses attention on important content.

Why it is used:

- remove unnecessary areas
- improve framing
- focus on important subject
- reduce image size

Example:

Cropping a photo to remove unwanted background objects.

Real-life use:

Photography, social media posts, banners.

2. Clip to Object

Clip to Object places an image inside a specific shape or object boundary so that only the part inside the shape remains visible.

This creates creative and professional design effects.

Why it is used:

- create custom image shapes
- improve visual design
- make layouts more attractive

Example:

Placing a photo inside a circular frame for a profile design.

Real-life use:

Profile cards, advertisements, presentations.

Colour Systems

1. RGB (Red, Green, Blue)

RGB is a colour system used for **digital screens** such as computers, mobile phones, televisions, and projectors. It produces colours by combining **Red, Green, and Blue light** in different intensities. This is an **additive colour model**, meaning colours are created by adding light.

How it produces colour:

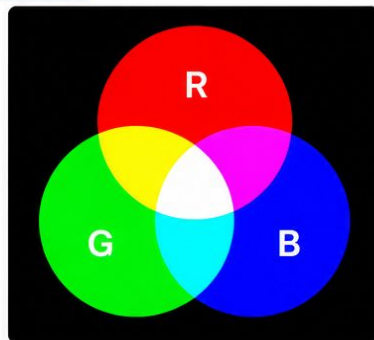
- Red + Green = Yellow
- Red + Blue = Magenta
- Green + Blue = Cyan
- Red + Green + Blue (full intensity) = White
- No light = Black

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How it produces colour:

Red + Green = Yellow



Red + Blue = Magenta



Green + Blue = Cyan



Red + Green + Blue
(full intensity) = White



No light = Black



RGB Intensity Example 0%



RGB Values

R: 0	R: 255	R: 255	R: 255	R: 0	R: 0	R: 0	R: 128	R: 255	R: 255
G: 0	G: 0	G: 128	G: 255	G: 255	G: 255	G: 0	G: 0	G: 0	G: 255
B: 0	B: 0	B: 0	B: 0	B: 0	B: 255	B: 255	B: 255	B: 255	B: 255

Properties:

- light-based colour system
- bright and vibrant colours
- wide colour range
- suitable for digital displays

Purpose:

Used to display images on electronic screens.

Uses:

- websites
- digital posters
- presentations
- mobile applications
- video graphics

Example:

A YouTube thumbnail viewed on a laptop screen uses RGB colours.

2. HSL (Hue, Saturation, Lightness)

Explanation:

HSL is a colour model that represents colour in a more human-friendly way.

It allows designers to control colour based on appearance rather than technical RGB values.

Components:

Hue: Base colour (red, blue, green, etc.)

Saturation: Colour intensity or richness

Lightness: Brightness level of the colour

How it produces colour:

Colour is created by adjusting:

- hue angle
- saturation percentage
- lightness percentage

Example:

Hue = Blue

High saturation = strong blue

Low saturation = dull grey-blue

High lightness = lighter blue

Properties:

- intuitive colour control
- easy colour adjustment
- useful for editing brightness and intensity

Purpose:

Makes colour editing easier for designers.

Uses:

- photo editing
- digital painting
- UI design
- graphic design software

Example:

Changing a bright red button to a softer pink using saturation/lightness controls.

3. CMYK (Cyan, Magenta, Yellow, Black)

CMYK is used for **printing**.

It produces colours by combining coloured inks instead of light.

This is a **subtractive colour model**, meaning colours are created by absorbing light.

How it produces colour:

- Cyan + Magenta + Yellow combine to create darker colours
- Black (K) is added for true dark tones and sharp text

More ink = darker colour

Properties:

- print-based colour system
- less vibrant than RGB
- accurate for physical printing
- ink dependent

Purpose:

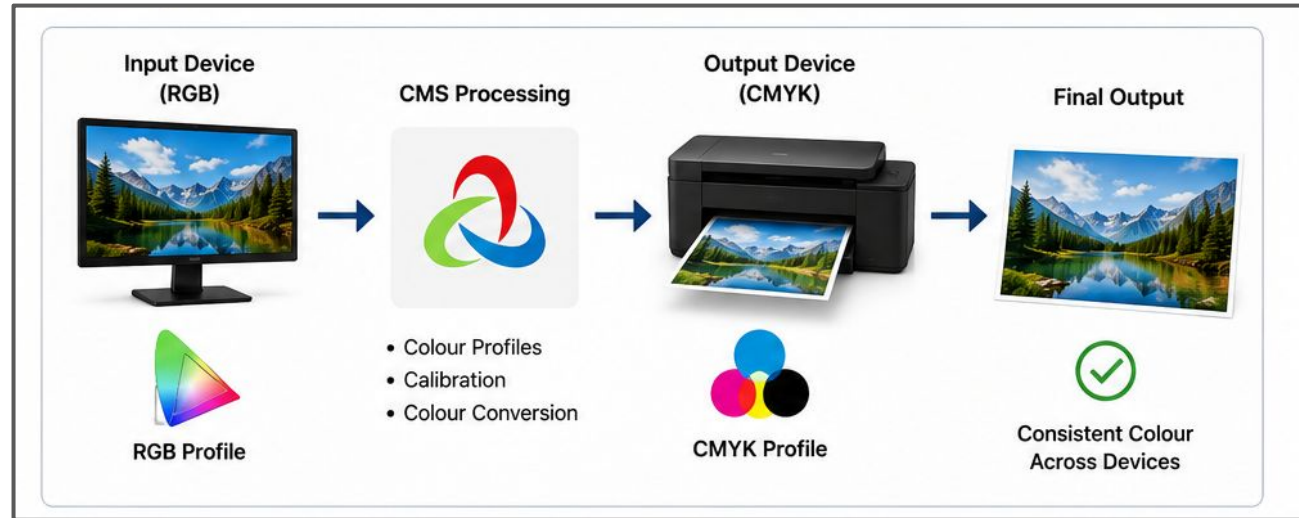
Used for printed output.

Uses:

- brochures
- magazines
- business cards
- posters
- packaging

Example:

A printed school magazine uses CMYK colours.



4. CMS (Colour Matching System)

CMS stands for Colour Matching System.

It ensures colours remain consistent across different devices, printers, software, and output formats.

Without CMS, the same image may appear different on different devices.

How it produces colour:

CMS uses colour profiles and calibration rules to translate colours accurately between systems.

Example:

RGB screen colour → converted accurately for printer output.

Properties:

- colour consistency
- device calibration support
- accurate reproduction
- professional quality control

Purpose:

Ensures colour accuracy across platforms.

Uses:

- branding
- professional printing
- publishing
- product packaging

Example:

A company logo appears exactly the same on website, brochure, and printed banner.

Comparison Table

Colour System	Produces Colour Using	Main Use
RGB	Light (Red, Green, Blue)	Digital screens
HSL	Hue + Saturation + Lightness	Colour editing
CMYK	Ink (Cyan, Magenta, Yellow, Black)	Printing
CMS	Colour profiles & calibration	Colour consistency

Need for Different Image Resolutions

Image resolution refers to the amount of detail contained in an image. It is usually measured in **pixels (for digital images)** or **dots per inch (DPI) for printing**.

Higher resolution means more image detail and clarity, while lower resolution means less detail.

Different resolutions are required depending on whether the image is used for **screen display or printing**.

Why Different Image Resolutions are Needed

1. Screen Display

Images used on digital screens do not require extremely high resolution because screens display images using pixels.

Typical screen resolution:

72–150 PPI

Using appropriate resolution:

- loads faster
- saves storage space
- displays clearly on screens
- improves website/app performance

Example:

A website banner displayed on a laptop screen.

2. Printing

Printed images require much higher resolution because printing uses ink dots and fine detail.

Typical print resolution:

300 DPI or higher

Higher resolution ensures:

- sharp text
- clear images
- professional print quality
- accurate details

Example:

Magazine cover or poster printing.

Impact of Too Low Resolution

On Screen

If resolution is too low:

- blurry appearance
- pixelated image
- poor visual quality
- unprofessional design

Example:

A profile photo appears blurry when enlarged on a website.

In Print

If print resolution is too low:

- jagged edges
- unclear text
- poor image sharpness
- low print quality

Example:

A printed brochure logo looks blurry.

Impact of Too High Resolution

On Screen

If resolution is too high:

- unnecessarily large file size
- slower loading time
- increased storage usage
- reduced website performance

Example:

Uploading a 50 MB image to a webpage causes slow loading.

In Print

Very high resolution beyond required limits may:

- increase file size unnecessarily
- slow printing process
- consume more system resources

However, print quality may not improve noticeably beyond optimal resolution.

Comparison

Usage	Recommended Resolution	Effect
Screen display	72–150 PPI	Fast loading, clear display
Print	300 DPI+	Sharp professional printing

Bitmap and Vector File Formats

1. Bitmap File Formats

Explanation:

Bitmap graphics are made up of tiny dots called **pixels**. Each pixel stores colour information, and together they form the complete image.

Bitmap images are resolution-dependent, meaning enlarging them too much causes pixelation and loss of quality.

They are best suited for **photographs and detailed images**.

Common Bitmap File Formats

JPEG / JPG (Joint Photographic Experts Group)

A widely used bitmap format that compresses image size.

Properties:

- small file size
- lossy compression
- good for detailed images
- supports millions of colours

Suitable for:

- digital photographs
- websites
- social media images

Example Scenario:

Uploading a product photo to an online shopping website.

PNG (Portable Network Graphics)

A bitmap format that supports transparency.

Properties:

- lossless compression
- transparent background support
- high image quality

Suitable for:

- logos with transparent background
- web graphics
- icons

Example Scenario:

Using a company logo on a website with transparent background.

GIF (Graphics Interchange Format)

Bitmap format supporting simple animation.

Properties:

- limited colours
- small file size
- animation support

Suitable for:

- simple animations
- icons
- web graphics

Example Scenario:

Animated loading icon on a website.

TIFF (Tagged Image File Format)

High-quality bitmap format often used professionally.

Properties:

- very high quality
- large file size
- suitable for printing

Suitable for:

- professional printing
- publishing
- scanned images

Example Scenario:

Printing a magazine advertisement.

BMP (Bitmap Image File)

Standard uncompressed bitmap format.

Properties:

- very large file size
- high quality
- minimal compression

Suitable for:

- basic image storage
- editing purposes

Example Scenario:

Saving raw edited graphics.

2. Vector File Formats

Explanation:

Vector graphics are created using mathematical paths, lines, curves, and shapes instead of pixels.

They are resolution-independent, meaning they can be resized without losing quality.

Best suited for graphics requiring scaling.

Common Vector File Formats

SVG (Scalable Vector Graphics)

A vector format designed for digital graphics.

Properties:

- scalable without quality loss
- lightweight
- editable

Suitable for:

- website logos
- icons
- UI graphics

Example Scenario:

Website logo that must display clearly on all screen sizes.

PDF (Portable Document Format)

A format that can contain vector graphics, bitmap images, and text.

Features:

- Preserves layout
- Print-friendly
- Widely supported

Suitable for:

- Posters
- Brochures
- Documents
- Print-ready designs

Scenario Example:

Sharing a brochure design with a printing company.

Bitmap vs Vector Comparison

Feature	Bitmap	Vector
Made of	Pixels	Shapes/Lines
Quality when resized	Decreases	Remains same
File size	Larger	Usually smaller
Best for	Photos	Logos, diagrams
Editing scalability	Limited	Excellent

Selection in Different Scenarios

Scenario	Best Format	Reason
Website photo	JPEG	Small size, fast loading
Transparent logo	PNG	Transparency support
Animated web image	GIF	Animation support
Professional print image	TIFF	High quality
Scalable logo	SVG	No quality loss when resized
Brochure/document sharing	PDF	Maintains layout

Exporting an Image in Different File Formats

Exporting an image means saving an image in a specific file format based on its intended use, such as web display, printing, editing, or sharing.

General Steps to Export an Image

- Open the image in image editing software (e.g., Photoshop, GIMP, Illustrator, Paint).
- Click **File** menu.
- Select **Export** or **Save As**.
- Choose the required file format.
- Adjust settings such as quality, compression, size, or transparency.
- Click **Save/Export**.

1) SVG (Scalable Vector Graphics)

- **How to Export:**

- Open vector artwork in Adobe Illustrator or Inkscape
- Click **File** → **Export As / Save As**
- Select **SVG (.svg)**
- Click **Save**

- **Used For:**

- Logos
- Icons
- Diagrams
- Web graphics

- **Advantages:**

- No quality loss when resized
- Small file size
- Editable format

- **Example:**

- Company logo for website

2) BMP (Bitmap)

- **How to Export:**
 - Open image in editing software
 - Click **File** → **Save As**
 - Select **BMP (.bmp)**
 - Click **Save**
- **Used For:**
 - Raw image storage
 - Windows applications
 - High-quality editing
- **Advantages:**
 - High image quality
 - No compression loss
 - Simple format
- **Example:**
 - Saving scanned images

3) JPG / JPEG

- **How to Export:**
 - Open image
 - Click **File** → **Export / Save As**
 - Select **JPEG (.jpg)**
 - Adjust quality settings
 - Click **Save**
- **Used For:**
 - Photographs
 - Social media
 - Website images
- **Advantages:**
 - Small file size
 - Fast loading
 - Good photo quality
- **Example:**
 - Uploading photos to website

4) PNG

- **How to Export:**
 - Open image
 - Click **File** → **Export As**
 - Select **PNG (.png)**
 - Enable transparency if needed
 - Save file
- **Used For:**
 - Transparent logos
 - Icons
 - Web graphics
- **Advantages:**
 - High quality
 - Supports transparency
 - Lossless compression
- **Example:**
 - Website logo with transparent background

5) GIF

- **How to Export:**
 - Open image or animation
 - Click **File** → **Export**
 - Select **GIF (.gif)**
 - Save animation settings
 - Click **Save**
- **Used For:**
 - Animated graphics
 - Loading icons
 - Memes
- **Advantages:**
 - Supports animation
 - Small file size
 - Web-friendly
- **Example:**
 - Animated loading symbol

6) TIF / TIFF

- **How to Export:**
 - Open image
 - Click **File** → **Save As / Export**
 - Select **TIFF (.tif)**
 - Choose quality settings
 - Save
- **Used For:**
 - Professional printing
 - Publishing
 - High-quality photography
- **Advantages:**
 - Excellent quality
 - Lossless format
 - Best for print
- **Example:**
 - Magazine/poster printing

7) PDF

- **How to Export:**
 - Open image/design
 - Click **File** → **Save As / Export**
 - Select **PDF (.pdf)**
 - Choose print/sharing settings
 - Save file
- **Used For:**
 - Documents
 - Brochures
 - Posters
 - Print-ready files
- **Advantages:**
 - Preserves layout
 - Easy sharing
 - Supports images and vector graphics
- **Example:**
 - Brochure for printing

Changing Solid and Gradient Fills

Changing fills refers to modifying the color appearance of an object, shape, text, or design element to improve its visual look.

Solid Fill:

A solid fill applies a single uniform color to the entire object without any variation. It gives a clean and simple appearance.

Gradient Fill:

A gradient fill applies a gradual transition between two or more colors, creating a blended effect. It gives depth and a more attractive visual appearance.

How to Change Solid or Gradient Fill:

Select the object or shape, open the formatting or fill options, choose either solid fill or gradient fill, select the required color or color combination, and apply the changes.

Uses:

Solid fills are commonly used for buttons, shapes, text backgrounds, and simple designs. Gradient fills are used for banners, posters, logos, website backgrounds, and creative designs.

Advantages:

Solid fills provide clarity, simplicity, and easy readability. Gradient fills make the design visually appealing, add depth, and create a professional look.

Example:

Using a solid blue color for a button in a website design or applying a pink-to-purple gradient for a poster background.

Assessment

Multiple Choice Questions

1. Which colour model is mainly used for printing?

- A) RGB
- B) HSL
- C) CMYK
- D) GIF

2. Which file format is best for a transparent website logo?

- A) JPEG
- B) PNG
- C) BMP
- D) TIFF

3. Which tool is used to arrange objects one above another?

- A) Crop
- B) Order
- C) Colour Picker
- D) Grid

4. Which transform tool is used to turn an object around a fixed point?

- A) Resize
- B) Rotate
- C) Move
- D) Crop

5. What happens if image resolution is too low for printing?

- A) Image becomes sharper
- B) File size decreases with better quality
- C) Image appears blurry or pixelated
- D) Colours become brighter

Case Study 1 — "The School Magazine Cover"

Background

Priya is a Year 12 student tasked with designing the front cover of her school's annual magazine. The cover must:

- Feature the school logo (to be reused at various sizes across the magazine)
- Include a photograph of the winning sports team
- Have a decorative background with a gradient
- Be sent to a professional printing company for 500 physical copies
- Also be published on the school website

Priya is using a professional graphics application with full layer, transform, colour, and export support.

Case Study Q1. Priya wants to place the school logo, the photograph, and the background gradient on separate layers so she can edit them independently.

- (a) Explain why working with separate layers is beneficial in this design.
- (b) Priya accidentally edits the photograph layer when she meant to edit the logo layer. Describe TWO layer features she could have used to prevent this mistake.

Case Study Q2. The school logo needs to appear on the magazine cover, on bookmarks (5 cm × 15 cm), and on a pull-up banner (85 cm × 200 cm).

- (a) Recommend the most suitable file format for the school logo. Justify your answer.
- (b) The team photograph was taken at 72 PPI. Explain the problem this may cause when the magazine is professionally printed, and state the resolution that should have been used.

Case Study Q3. Priya needs to export two versions of the finished magazine cover — one for print and one for the school website.

State the most appropriate file format for each version and give ONE reason for each choice.

Case Study 2 — "The Brand Identity Project"

Background

A small startup company called *NovaBrew* (a coffee brand) has hired a freelance graphic designer named Rohan. Rohan must create:

- A circular logo using geometric shapes
- A colour scheme consistent with their brand across web and print materials
- Social media post graphics (to be posted as still images on Instagram)
- A promotional flyer to be printed and distributed locally

Rohan uses a vector graphics application for the logo and a raster application for the promotional flyer.

Case Study Q1. To create the NovaBrew circular logo, Rohan draws a large circle and a smaller inner circle. He wants to create a ring shape (like a donut) by removing the inner circle from the large circle.

(a) Name the Boolean grouping operation Rohan should use and explain how it works.

(b) After creating the ring, Rohan adds text inside it. He selects both the ring and the text and applies the Group function. State TWO benefits of grouping these objects together.

Case Study Q2. NovaBrew's brand colour is a deep brown. Rohan's client approves the colour on screen, but when the promotional flyer is printed, the colour looks significantly different — much duller and less warm.

(a) Explain why this colour difference occurs between screen and print.

(b) Describe how a Colour Management System (CMS) could help Rohan avoid this problem in the future.

Case Study Q3. Rohan is exporting the Instagram post graphic and the promotional flyer for print. The Instagram post features the logo on a transparent background so it can be overlaid on different coloured backgrounds by the social media team.

(a) Recommend a suitable file format for the Instagram post. Justify your choice with reference to TWO properties of the format.

(b) State ONE format Rohan should NOT use for the Instagram post and explain why it is unsuitable.